Mahesh Ponnam

Data Scientist

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Objective

Accomplished Data Scientist with 4+ **years** of experience in leveraging data analytics, machine learning, and statistical modeling to drive business insights and decision-making. Skilled in managing end-to-end data science projects, from data collection and preprocessing to model development and deployment.

Profile Summary

- Highly experienced **Data Scientist** with over **4+ years**' experience in **Data Extraction, Data Modelling, Data Wrangling, Statistical Modeling, Data Mining, Machine Learning** and **Data Visualization.**
- Expertise in transforming business resources and requirements into manageable data formats and analytical models, designing algorithms, building models, developing **data mining** and **reporting solutions** that scale across a massive volume of **structured and unstructured data.**
- Proficient in managing entire **data science project** life cycle and actively involved in all the phases of project life cycle including data acquisition, data cleaning, data engineering, features scaling, features engineering, statistical modeling, testing and validation and data visualization.
- Proficient in **Machine Learning algorithm** and **PredictiveModeling** including **RegressionModels**, **Decision Tree, Random Forests, Sentiment Analysis, Naïve Bayes Classifier, SVM, Ensemble Models.**
- Proficient in **Statistical Methodologies** including **Hypothetical Testing, ANOVA, Time Series, Principal Component Analysis, Factor Analysis, Cluster Analysis, Discriminant Analysis.**
- Knowledge on time series analysis using **AR, MA, ARIMA, GARCH** and **ARCH model.**
- Knowledge on **Natural Language Processing (NLP) algorithm** and **Text Mining.**
- Worked in large scale database environment like **Hadoop** and **MapReduce**, with working mechanism of **Hadoop clusters,nodes** and **Hadoop Distributed File System (HDFS).**
- Strong experience with **Python (2.x,3.x)** to develop analytic models and solutions.
- Proficient in **Python 2.x/3.x with SciPy Stack packages including NumPy, Pandas, SciPy, Matplotlib** and **Python.**
- Working experience in **Hadoop ecosystem and Apache Spark framework** such as **HDFS, MapReduce, HiveQL, SparkSQL, PySpark.**
- Very good experience and knowledge in provisioning virtual clusters under **AWS cloud** which includes services like **EC2**, **S3,** and **EMR.**
- Proficient in data visualization tools such as **Tableau, Python Matplotlib, R** Shiny to create visually **powerful** and actionable interactive reports and dashboards.
- Excellent **Tableau Developer,** expertise in building, publishing customized interactive reports and dashboards with customized parameters and user - filters using **Tableau (9.x/10.x).**
- Experienced in **Agile** methodology and **SCRUM** process.
- Strong business sense and abilities to communicate data insights to both technical and nontechnical clients.

Skills

Databases: MySQL, Postgre SQL, Oracle, HBase, Amazon Redshift, MS SQL Server 2016/2014/2012/2008 R2/2008, Taradata

Statistical Methods: Hypothetical Testing, ANOVA, Time Series, Confidence Intervals, Bayes Law, Principal Component Analysis (PCA), Dimensionality Reduction, Cross-Validation, Auto-correlation

Machine Learning: Regression analysis, Bayesian Method, Decision Tree, Random Forests, Support Vector Machine, Neural Network, Sentiment Analysis, K-Means Clustering, KNN and Ensemble Method, Natural Language Processing (NLP)

Hadoop Ecosystem: Hadoop 2.x, Spark 2.x, MapReduce, Hive, HDFS, Sqoop, Flume

Reporting Tools: Tableau Suite of Tools 10.x, 9.x, 8.x which includes Desktop, Server and Online, Server Reporting Services (SSRS) Data Visualization: Tableau, MatPlotLib, Seaborn, ggplot2

Languages: Python (2.x/3.x), R, SAS, SQL, T-SQL

Operating Systems: PowerShell, UNIX/UNIX Shell Scripting (via PuTTY client), Linux and Windows

Education

Masters from Wilmington University, USA

Work Experience

Client: Brown Brothers Harriman, Philadelphia, PA, USA **April 2024 - Present**
Role: Data Scientist

Description: Brown Brothers Harriman (BBH) is a renowned financial services firm offering investment management, private banking, and asset servicing solutions to institutional and individual clients. Developed predictive models and recommender systems using Python, Spark, and deep learning frameworks, applying machine learning, NLP, and clustering techniques to analyze customer behavior and optimize satisfaction.

Responsibilities:

- Collaborated with data engineers and operation team to implement **ETL** process, wrote and optimized **SQL queries** to perform data extraction to fit the analytical requirements.
- Performed data analysis by using Hive to retrieve the data from **Hadoop cluster, SQL** to retrieve data from RedShift.
- Explored and analyzed the customer specific features by using **Spark SQL.**
- Performed univariate and multivariate analysis on the data to identify any underlying pattern in the data and associations between the variables.
- Performed data imputation using **Scikit-learn package** in **Python.**
- Participated in features engineering such as feature intersection generating, feature normalize and label encoding with **Scikit-learn preprocessing.**
- Used **Python 3.X (numpy, scipy, pandas, scikit-learn, seaborn)** and **Spark 2.0 (PySpark, MLlib)** to develop variety of models and algorithms for analytic purposes.
- Developed and implemented predictive models using **machine learning algorithms** such as **linear regression, classification, multivariate regression, Naive Bayes, Random Forests, K-means clustering, KNN, PCA** and **regularization** for **data analysis.**
- Conducted analysis on assessing customer consuming behaviors and discover value of customers with **RMF analysis;** applied customer segmentation with clustering algorithms such as K**-Means Clustering** and **Hierarchical Clustering.**
- Built regression models include: **Lasso, Ridge, SVR, XGboost**to **predict Customer Life Time Value.**
- Built classification models include: **Logistic Regression, SVM, Decision Tree, Random Forest** to **predict Customer Churn Rate.**Used **numpy**, **scipy**, **pandas**, **nltk(Natural Language Processing Toolkit), matplotlib** to build the model.
- Formulated several graphs to show the performance of the students by demographics and their mean score in different USMLE exams.
- Application of various **Artificial Intelligence (AI)/machine learning algorithms** and **statistical modeling** like **decision** trees, **text analytics, natural language processing (NLP), supervised** and **unsupervised, regression models.**
- Used Principal Component Analysis in feature engineering to analyze high dimensional data.
- Performed **Data Cleaning**, features scaling, features engineering using pandas and **numpypackages** in **python** and build models using **deep learning frameworks.**
- Created deep learning models using **Tensorflow** and **keras** by combining all tests as a single normalized score and predict residency attainment of students.
- Used **XGB classifier** if the feature is an categorical variable and **XGBregressor** for continuous variables and combined it using **Feature** Union and **FunctionTransfomer** methods of **NaturalLanguage Processing.**
- Used F**-Score, AUC/ROC, Confusion Matrix, MAE, RMSE** to evaluate different Model performance.
- Designed and implemented recommender systems which utilized Collaborative filtering techniques to recommend course for different customers and deployed to **AWS EMR cluster.**
- Utilized natural language processing (**NLP**) techniques to Optimized Customer Satisfaction.
- Designed rich data visualizations to model data into human-readable form with **Tableau** and **Matplotlib**.

Environment: AWS RedShift, EC2, EMR, Hadoop Framework, S3, HDFS, Spark (Pyspark, MLlib, Spark SQL), Python 3.x (Scikit- Learn/Scipy/Numpy/Pandas/NLTK/Matplotlib/Seaborn), Tableau Desktop (9.x/10.x), Tableau Server (9.x/10.x), Machine Learning (Regressions, KNN, SVM, Decision Tree, Random Forest, XGboost, LightGBM, Collaborative filtering, Ensemble), NLP, Teradata, Git 2.x, Agile/SCRUM

Client: BNY Mellon, Pittsburgh, Pennsylvania, USA Jan 2024 - Mar 2024

Role: Data Scientist

Description: BNY Mellon is a global leader in investment and securities services, offering innovative solutions in asset management, wealth management, and treasury services to help clients achieve their financial goals.Developed predictive models and machine learning solutions for fraud detection and loan default prediction using Python, Spark, and AWS, applying advanced techniques like SMOTE, feature engineering, and ensemble methods for improved performance.

Responsibilities:

- Tackled highly imbalanced Fraud dataset using under sampling, oversampling with **SMOTE** and cost sensitive algorithms with **Python Scikit-learn.**
- Wrote complex **Spark SQL queries** for data analysis to meet business requirement.
- Developed **MapReduce/Spark Python** modules for predictive analytics &**machine learning** in **Hadoop** on **AWS.**
- Worked on data cleaning and ensured data quality, consistency, integrity using **Pandas**, **Numpy**.
- Participated in feature engineering such as feature intersection generating, feature normalize and label encoding with **Scikit- learn preprocessing.**
- Improved fraud prediction performance by using random forest and gradient boosting for feature selection with **Python-Scikit-learn.**
- Performed **Naïve Bayes, KNN, Logistic Regression, Random Forest, SVM and XGboost** to identify whether a loan will default or not.

Client: (DXC Technology) Tops Markets, Mumbai, India Oct 2020 - Nov 2022

Role: Data Analyst/ Data Scientist

Description: Tops Markets is a leading supermarket retailer. Contributed to data analysis and modeling, developed and maintained detailed reports, and communicated insights to support strategic decision-making and project prioritization.

Responsibilities:

- Gathered, analyzed, documented and translated application requirements into data models and Supports standardization of documentation and the adoption of standards and practices related to data and applications.
- Participated in **Data Acquisition** with **Data Engineer** team to extract historical and real-time data by using **Sqoop, Pig, Flume, Hive, MapReduce** and **HDFS.**
- Wrote user defined functions (**UDFs**) in **Hive** to manipulate strings, dates and other data.
- Performed **Data Cleaning, features scaling, features** engineering using pandas and **numpypackages** in **python**.
- Applied clustering algorithms i.e. **Hierarchical, K-means using Scikit** and**Scipy.**
- Performs complex pattern recognition of automotive time series data and forecast demand through the **ARMA** and **ARIMA** models and exponential smoothening for multivariate time series data.
- Delivered and communicated research results, recommendations, opportunities to the managerial and executive teams, and implemented the techniques for priority projects.
- Designed, developed and maintained daily and monthly summary, trending and benchmark reports repository in **Tableau** Desktop.
- Generated complex calculated fields and parameters, toggled and global filters, dynamic sets, groups, actions, custom color palettes, statistical analysis to meet business requirements.
- Implemented visualizations and views like combo **charts**, **stacked bar charts, pareto charts, donut charts, geographic-maps, spark lines, crosstabs** etc.
- Published workbooks and extract data sources to Tableau Server, implemented row-level security and scheduled automatic extract refresh.

Environment: Machine learning (KNN, Clustering, Regressions, Random Forest, SVM, Ensemble), Linux, Python 2.x (Scikit- Learn/Scipy/Numpy/Pandas), R, Tableau (Desktop 8.x/Server 8.x), Hadoop, Map Reduce, HDFS, Hive, Pig, HBase, Sqoop, Flume, Oracle 11g, SQL Server 2012

Client: Johnson & Johnson, Mumbai, India **Aug 2019 - Sep 2020**

Role: Data Analyst

Description: Johnson & Johnson is a global healthcare leader known for its innovative products across pharmaceuticals, medical devices, and consumer health sectors. Involved in developing and optimizing data integration processes, creating detailed financial reports, and performing advanced statistical analyses to support business decision-making.

Responsibilities:

- Used **SSIS** to create **ETL packages** to Validate, Extract, Transform and Load data into **Data Warehouse** and **Data Mart.**
- Maintained and developed complex **SQL queries**, **stored procedures, views, functions,** and reports that meet customer requirements using **Microsoft SQL Server 2008 R2.**
- Created **Views** and **Table-valued Functions, Common Table Expression** (**CTE**), **joins**, complex subqueries to provide the reporting solutions.
- Optimized the performance of queries with modification in **T-SQL queries**, removed the unnecessary columns and redundant data, normalized tables, established joins and created index.
- Created **SSIS packages** using **Pivot Transformation, Fuzzy Lookup, Derived Columns, Condition Split, Aggregate, Execute SQL Task, Data Flow Task** and **Execute Package Task.**
- Migrated data from **SAS** environment to **SQL Server 2008** via **SQL Integration Services (SSIS).**
- Developed and implemented several types of **Financial Reports (Income Statement, Profit& Loss Statement, EBIT, ROIC Reports)** by using **SSRS.**
- Developed parameterized dynamic performance **Reports** (**Gross Margin, Revenue base on geographic regions, Profitability based on web sales** and **smartphone app sales**) and ran the reports every month and distributed them to respective departments through mailing server subscriptions and **SharePoint server.**
- Designed and developed new reports and maintained existing reports using **Microsoft SQL Reporting Services (SSRS)** and **Microsoft Excel** to support the firm's strategy and management.
- Created sub-reports, drill down reports, summary reports, parameterized reports, and ad-hoc reports using **SSRS**.
- Used **SAS/SQL** to pull data out from databases and aggregate to provide detailed reporting based on the user requirements.
- Used **SAS** for pre-processing data**, SQL queries**, **data analysis, generating reports, graphics, and statistical analyses.**
- Provided statistical research analyses and data modeling support for mortgage product.
- Perform analyses such as regression analysis, logistic regression, discriminant analysis, cluster analysis using **SAS** programming.

Environment: SQL Server 2008 R2, DB2, Oracle, SQL Server Management Studio, SAS/ BASE, SAS/SQL, SAS/Enterprise Guide, MS BI Suite (SSIS/SSRS), T-SQL, SharePoint 2010, Visual Studio 2010, Agile/SCRUM